# Yan Study OS - One-Page App Summary

## What it is

Yan Study OS is a personal IB Diploma study command center built with Next.js and TypeScript.
It combines exam planning, timed study logging, and todo management in one authenticated dashboard.

## Who it is for

Primary persona: Yan, an IBDP Year 2 student preparing for final exams.

## What it does

- Passcode login with server-side verification and signed cookie sessions.
- Shows nearest upcoming exams with live countdown and subject-colored progress bars.
- Renders a 120-day forward calendar with exam-day highlights and day study intensity.
- Includes a subject-based study timer (start, pause/resume, end) with auto-resume after refresh.
- Logs study sessions and aggregates cumulative time totals by subject.
- Manages todos with deadlines, ultra-priority sorting, and archive/restore.
- Queues offline mutations in API mode and auto-syncs when connectivity returns.

## How it works (repo-evidenced architecture)

- UI: src/app/page.tsx renders StudyDashboard as a client component.
- Data adapter: src/lib/client/repository.ts switches local (localStorage) vs api mode via NEXT_PUBLIC_DATA_MODE.
- Auth/API: /api/auth/* handles login/logout/session; protected /api/* routes enforce requireSession.
- Store selector: src/lib/server/store/index.ts uses DATA_BACKEND to choose local-file or DynamoDB store.
- Local store writes runtime-data/yan-study-os.json and an exams backup file.
- Dynamo store reads/writes sessions, todos, and settings tables from YAN_STUDY_*_TABLE env vars.
- Infra: amplify/ defines Amplify Gen 2 data models and DynamoDB table provisioning.
- Additional external services (analytics/monitoring/jobs): Not found in repo.

## How to run (minimal)

1. npm install
2. cp .env.example .env.local
3. node scripts/generate-passcode-hash.mjs "your-passcode"
4. In .env.local set YAN_STUDY_PASSCODE_HASH and SESSION_SECRET
5. npm run dev, then open http://localhost:3000